

LEMISON POWDER 100MG/G

Ingredient(s)

Each gram contains:

Levamisole Hydrochloride 100mg
(eq. to Levamisole 84.84mg)

Benzethonium chloride (as preservative) 0.04mg

Pharmacodynamics

Levamisole stimulates the parasympathetic and sympathetic ganglia in susceptible worms. At higher levels, levamisole interferes with nematode carbohydrate metabolism by blocking fumarate reduction and succinate oxidation. The net effect is a paralyzing effect on the worm which is then expelled alive. Levamisole's effects are considered to be nicotine-like in action.

Levamisole's mechanism of action for its immuno-stimulating effects is not well understood. It is believed it restores cell-mediated immune function in peripheral T-lymphocytes and stimulates phagocytosis by monocytes. Its immune stimulating effects appear to be more pronounced in animals that are immune-compromised.

Pharmacokinetics

Levamisole is absorbed from the gut after oral dosing and through the skin after dermal application, although bio-availabilities are variable. It is reportedly distributed throughout the body. Levamisole is primarily metabolized with less than 6% excreted unchanged in the urine. Approximately 40% is excreted in urine in 12 hours. Thereafter, urinary excretion decreases and only another 8% is eliminated over the next 8 days. Elimination in feces over an 8-day period accounts for approximately 41% of the dose, the bulk of which passes in 12–24 hours. A small amount is expelled in respired gases (i.e., 0.2% of the dose during a 48-hour period immediately following dosing). Approximately 0.9% of the initial dose is found in tissues (principally degradative and excretion organs like the liver and kidney) at 12–24 hours after dosing. Plasma elimination half-lives have been determined for several veterinary species: Cattle 4–6 hours; Dogs 1.8–4 hours; and Swine 3.5–6.8 hours. Metabolites are excreted in both the urine (primarily) and feces.

Indication

A broad spectrum anthelmintic for use in the treatment and control of nematode infections.

Swine: Large roundworms (*Ascaris suum*), nodular worms (*Oesophagostomum* species), intestinal thread worms (*Strongyloides ransomi*), and lungworms (*Metastrongylus* species).

Pigs maintained under conditions of constant exposure to worms may require retreatment within 4 to 5 weeks after the first treatment. Consult your veterinarian before administering to sick swine.

Cattle: Stomach worms (*Haemonchus placei*, *Ostertagia ostertagi*, *Trichostrongylus axei*); intestinal worms (*T. longispicularis*, *Cooperia oncophora*, *C. punctata*, *Nematodirus spathiger*, *Bunostomum phlebotomum*, *Oesophagostomum radiatum*); and lungworms (*Dictyocaulus viviparus*).

Conditions of constant helminth exposure may require retreatment 2 to 4 weeks after the first treatment. Consult your veterinarian before using in severely debilitated animals.

Sheep: Stomach worms (*Haemonchus contortus*, *Trichostrongylus axei*, *Teladorsagia circumcincta*), intestinal worms (*Trichostrongylus colubriformis*, *Cooperia curticei*, *Nematodirus spathiger*, *Bunostomum trigonocephalum*, *Oesophagostomum columbianum*, *Chabertia ovina*) and lungworms (*Dictyocaulus filaria*).

Conditions of constant helminth exposure may require retreatment 2 to 4 weeks after the first treatment. Consult veterinarian before using in severely debilitated animals.

Target Species

Swine (no use class stated or implied)

Cattle (Cattle, beef, dairy, not breeding age)

Sheep (DOMESTIC) (Sheep, no use class stated or implied)

Recommended Dose

Swine, cattle, sheep: 8g Lemison Powder per 100kg body weight mix in drinking water.

Mode of Administration

Orally via drinking water.

Contraindication

Contraindicated in lactating animals.

Warning and Precautions

1. In animals that are severely debilitated, or have significant hepatic or renal impairment. Delay use in cattle that are stressed due to vaccination, dehorning or castration.
2. Symptoms of levamisole toxicity often mimic those of organophosphate toxicity. Symptoms may include hypersalivation, hyperesthesias and irritability, chronic seizures, CNS depression, dyspnea, defecation, urination, and collapse.
3. Acute levamisole overdosage can result in death due to respiratory failure.
4. Cardiac arrhythmias may also be seen.
5. Muzzle foam or salivation may be observed. However, the reaction will disappear with a few hours after medication.
6. If pigs are infected with mature lungworms, coughing and vomiting may be observed soon after medication. The reaction is due to the expulsion of worms from the lungs and will be over in several hours.

Interaction with Other Medicaments

1. Other nicotine-like compounds (e.g., pyrantel, morantel, diethylcarbamazine), or cholinesterase-inhibitor drugs (e.g., organophosphates, neostigmine) could theoretically enhance the toxic effects of levamisole.
2. Levamisole may enhance the immune reaction and efficacy to Brucella vaccines.
3. Fatalities have been reported after concomitant levamisole and chloramphenicol administration.

Pregnancy and Lactation

1. Levamisole should not be used in lactating animals producing milk for human consumption.
2. There is no information regarding the safety in pregnant animals. Although levamisole is considered relatively safe to use in large animals that are pregnant, use only if the potential benefits outweigh the risks.

Side Effects

- 1) In cattle, muzzle-foaming or hypersalivation, excitement or trembling, lip-licking and head shaking. These effects are generally noted with higher than recommended doses or if levamisole is used concomitantly with organophosphates. Symptoms generally subside within 2 hours.
- 2) In sheep, levamisole may cause a transient excitability after dosing.
- 3) In goats, levamisole may cause depression, hyperesthesia and salivation.
- 4) In swine, levamisole may cause salivation or muzzle foaming. Swine infected with lungworms may develop coughing or vomiting.
- 5) In dogs, adverse effects may include GI disturbances (usually vomiting, diarrhea), neurotoxicity (panting, shaking, agitation or other behavioral changes), agranulocytosis, dyspnea, pulmonary edema, immune-mediated skin eruptions (erythroedema, erythema multiforme, toxic epidermal necrolysis) and lethargy.
- 6) In cats, adverse effects include hypersalivation, excitement, mydriasis and vomiting.

Environmental Property

None has been mentioned in animal.

Symptoms and Treatment of Overdose

Emptying the gut and/or administering charcoal with cathartics may be indicated

Storage Condition

Store at temperature below 30°C. Protect from direct sunlight and water.

Shelf Life

4 years.

Withdrawal Periods

Swine, sheep: Do not slaughter for food within 72 hours of treatment

Cattle: Do not slaughter for food within 48 hours of treatment. Not for use in dairy animals of breeding age.

Maximum Residual Limit (MRL)

Swine, cattle and sheep = 10µg/kg (muscle, kidney and fat).

Product Description

A white color powder, and without other foreign matter.

1kg aluminum pouch.



® Made by YSP
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